

KŌURA POPULATIONS IN LAKE TAUPŌ AND COMMENTS ON THE EFFECTS OF CATFISH



REPORT PREPARED FOR NGĀTI TŪWHARETOA MĀORI TRUST BOARD

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Kōura populations in Lake Taupō and comments on the effects of catfish

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Cover image

Typical kōura catch from one whakaweku using the tau kōura method, Hiruharama Bay, Lake Taupō, 17 December 2014.

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1. INTRODUCTION

Kōura or freshwater crayfish (*Paranephrops planifrons*) support an important customary fishery in Lake Taupō and are considered a taonga species by Ngāti Tūwharetoa. Historically, kōura were plentiful in Lake Taupō and large quantities were harvested for consumption and trading (Fletcher 1919). Anecdotal evidence suggests that kōura populations have declined over the last 100 years due to introductions of exotic fish and plant species. Exotic fish prey on kōura while introduced macrophytes can hinder kōura movement and accumulate large amounts of organic detritus. The lack of quantitative information on trends in kōura abundance makes it difficult for Ngāti Tūwharetoa and government agencies to manage this taonga species. Until recently the main reason for the lack of information on kōura in Lake Taupō and New Zealand lakes has been the absence of a suitable representative sampling method. However, the development of the traditional tau kōura method (Fig. 1) for research (Kusabs, et al. 2005; Kusabs and Quinn 2009; Kusabs, et al. 2015a; Kusabs, et al. 2015b) and monitoring (Kusabs and Butterworth 2013; Kusabs, et al. 2014) has greatly increased the understanding of kōura populations in the Te Arawa Lakes.

The tau kōura was used in pre-European times by Ngāti Tūwharetoa iwi to harvest kōura and koaro (*Galaxias brevipinnis*) from Lake Taupō (Fletcher 1919). This method involved resting whakaweku (bundles of bracken fern fronds, *Pteridium esculentum*) on the lakebed for kōura to take refuge in, and then retrieving the whakaweku into a boat to harvest the kōura (Fig. 1). Today the most popular method of harvesting kōura is by rama kōura, whereby relatively small quantities of kōura are collected from the shallow littoral at night using a torch and a hand-held scoop net.

Brown bullhead catfish (*Ameiurus nebulosus*) (hereafter catfish) were first discovered in Lake Taupō during the early 1980s. It is unclear how they arrived in Lake Taupō, although McDowall (1990) believed that they originated from an illegal liberation into the southern end of the lake. A native of the southern and eastern states of America, these catfish have been in New Zealand since 1878 and are now widespread throughout the Waikato region. The effects of catfish on the renowned rainbow trout fishery are of concern to trout fishermen and the Fishery Manager - the Department of Conservation. Although, catfish have been reported to commonly consume kōura in Lake Taupō (Barnes and Hicks 2003) their impact on kōura populations is unknown as there is no quantitative information on kōura abundance prior to the introduction of catfish to the lake. To obtain some insight into the effects of catfish on kōura we compared population characteristics of kōura in Lake Taupō with those from deep lakes in the neighbouring Te Arawa Lakes District where catfish are not known to be present.

2. Objectives

The aims of this study were to improve the basis for sustainable management of kōura in Lake Taupō by characterising population abundance, biomass and size distribution, and evaluating the effects of catfish by comparing Lake Taupō kōura with those from lakes Rotoiti, Rotomā and Tarawera, where catfish are absent.

3. Methods

3.1 Study Area

Lake Taupō, New Zealand's largest lake has an area of 616 km² and a natural catchment area of 286,515 ha as well as drainage from a further 67,110 ha via the artificially diverted Tongariro hydro-electricity scheme (Barnes 1996). Lake Taupō is warm and monomictic, stratifying between November and May to a minimum depth of 25–35 m (Vincent and Forsyth 1983). Peak phytoplankton biomass is during winter (Jolly 1968; Viner and White 1987) with low nutrient concentrations limiting phytoplankton growth during summer (Vincent and Forsyth 1983).

Most of the eastern side of the lake is wind exposed with sandy substrate and little plant growth. In more sheltered areas, large communities of macrophytes can dominate. Native macrophytes have largely been smothered by the introduced macrophytes *Elodea canadensis*, *Lagarosiphon major* and *Ceratophyllum demersum*. There are three species of native fish in Lake Taupō, koaro (*Galaxias brevipinnis*), common bully (*Gobiomorphus cotidianus*) and common smelt (*Retropinna retropinna*). Introduced fish species include brown trout (*Salmo trutta*), rainbow trout (*Oncorhynchus mykiss*) and catfish.

3.2 Tau kōura location and lay out

Kōura were sampled using the tau kōura, a traditional Te Arawa/Ngāti Tūwharetoa method of harvesting kōura in the Rotorua and Taupō lakes (Fig. 1). The methods used in this study are described in detail by Kusabs and Quinn (2009). Three tau kōura were deployed on 22 October 2014 at three locations in Lake Taupō, Hiruharama, Motuoapa and Waihi (Table 1, Fig. 2). Each tau kōura was comprised of 10 dried bracken fern (*Pteridium esculentum*) bundles, with c. 10-14 dried fronds per bundle, which were attached to a bottom line (a 250 m length of sinking anchor rope) and set at Hiruharama, Motuoapa and Waihi in depths ranging from 9 to 27 m, 18 to 33 m and 17 to 33 m, respectively (Fig. 2). The tau kōura were set on 22 October 2014 and left for approximately six weeks to allow kōura to colonise the fern and retrieved on 17 December 2014 and 24 February 2015. The tau kōura were placed back into the water once kōura had been monitored.

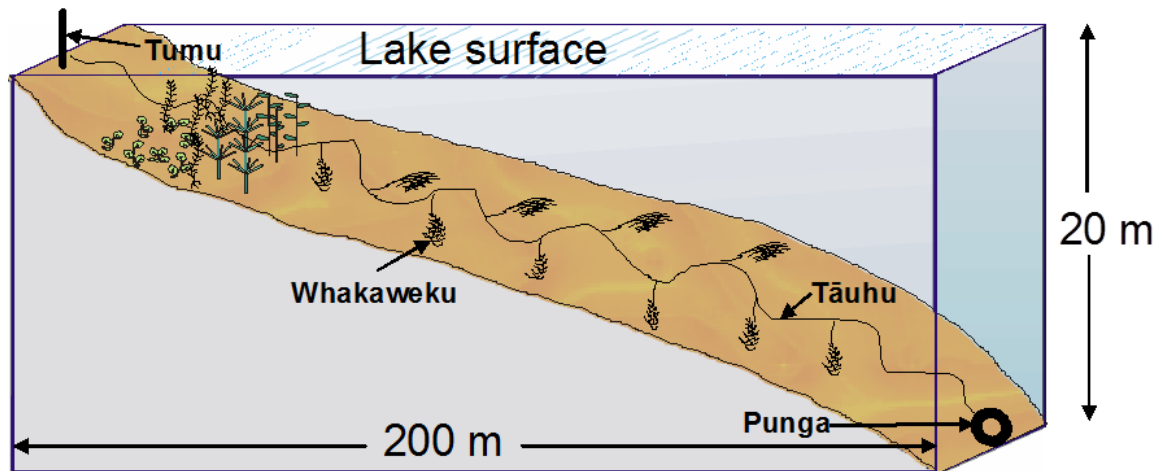


Figure 1 Schematic diagram of a tau kōura. The depth and length of tau are indicative and can be varied depending on lake bathymetry.

Table 1 Tau kōura monitoring sites, Lake Taupō, October 2014. GPS coordinates in NZ Transverse Mercator Projection (NZTM).

Site	Northing	Easting
Hiruharama Bay	1863128.8	5709574.8
Motuoapa Bay	1848512.4	5688197.5
Waihi Bay	1838098.3	5686022.3

3.3 Kōura measurements

Orbit-carapace length (OCL, mm; Fig. 3) of each kōura was measured using vernier callipers (± 0.5 mm) and the sex of kōura (OCL > 11 mm) assessed. A power regression equation (previously determined by B. Hicks and P. Riordan, University of Waikato) was used to determine kōura wet weight (g) from OCL (mm):

$$\text{Wet weight} = 0.000648 \text{ OCL}^{3.0743}$$

After processing, all kōura were returned to the water in close proximity to the tau kōura. Catch per unit effort (CPUE) was defined as the number of kōura per whakaweku and biomass per unit effort (BPUE) as estimated wet weight (g) of kōura per whakaweku.

3.4 Data analysis

Differences between mean CPUE, BPUE and OCL of kōura in lakes Taupō, Rotoiti, Rotomā and Tarawera were assessed using Kruskal-Wallis analysis of variance (ANOVA) by ranks in R version 3.0.0. Kruskal-Wallis is a non-parametric method for testing whether samples originate from the same distribution, it is used to compare two or more means that are independent, and that may have different sample sizes.



Figure 2. Map of Lake Taupō showing location of kōura and catfish monitoring sites. 2008 = Kusabs (2008), 2014 = 2014/15 sampling sites.

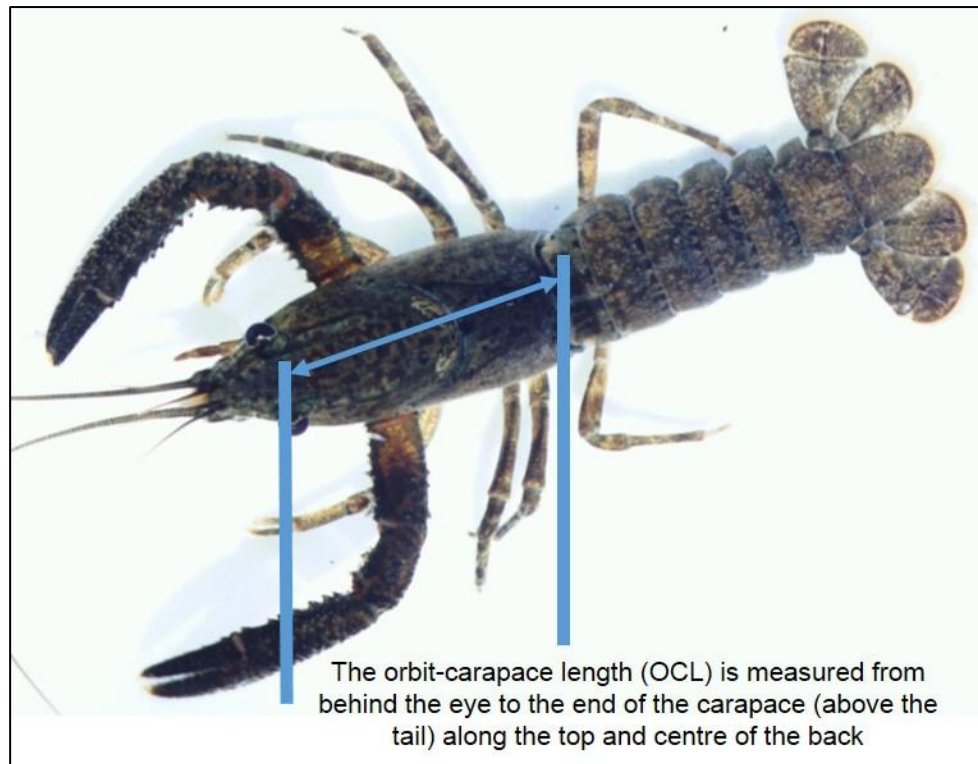


Figure 3. The orbit-carapace length (OCL) is measured from behind the eye to the end of the carapace along the top centre of the back.



Figure 4. Rakei Taiaroa and Anthony Karaitiana constructing whakaweku (bracken fern bundles) each composed of 10-12 bracken fern fronds, 22 October 2014.

3.5 Catfish

The Department of Conservation has monitored the CPUE and size of catfish (Fig. 5) in Lake Taupō on a semi-regular basis from 1996 to 2015 (Kusabs and Taiaroa 2015). The main aim of this on-going monitoring programme is to “identify long-term trends in the catfish population”. Sampling of catfish is carried out at three sites, Waihi, Motuoapa and Whakaipo Bays (Fig. 2). Six fyke nets are set at Waihi and Motuoapa, and three at Whakaipo. The nets are set close to the shoreline in shallow water about 60 cm deep. Subsamples of the population are assessed for fork length (FL), weight and gender. Diet analysis has also been carried out on an irregular basis since 1999 (Department of Conservation records).



Figure 5 Catfish captured fly fishing in Lake Taupō. Photo courtesy of Kim Turia, Department of Conservation.

4 RESULTS

4.1 Tau kōura deployment

A complete survey of all three tau kōura was carried out in December 2014. Unfortunately, human interference resulted in the February surveys at Hiruharama (tau kōura removed/stolen) and at Motuoapa (tāuhu cut leaving only two whakaweku intact) meant that data was collected from all three sites in December 2014 and only Waihi Bay in February 2015.

4.2 Kōura abundance and biomass

A total of 1088 kōura were collected from tau kōura set at Hiruharama ($n = 234$), Motuoapa ($n = 409$) and Waihi ($n = 427$, 2 surveys) (Table 2). The highest mean CPUE for the December survey was recorded at Motuoapa (CPUE = 39.1 kōura per whakaweku) followed by Waihi (CPUE = 28.7 kōura per whakaweku) and Hiruharama (CPUE = 23.4 kōura per whakaweku) (Table 2, Fig. 6). At Waihi, kōura CPUE was higher in December 2014 (CPUE = 28.7 kōura per whakaweku) than in February 2015 (CPUE = 18.8 kōura per whakaweku).

The highest estimated mean biomass was also recorded at Motuoapa (BPUE = 663.4 g of kōura per whakaweku) followed by Hiruharama (BPUE = 622.4 g of kōura per whakaweku) and Waihi (BPUE = 386.7 g of kōura per whakaweku) (Table 2, Fig. 6). There were no significant differences in mean CPUE or BPUE between Hiruharama, Motuoapa and Waihi ($p > 0.5$).

Table 2 Sampling site, date, number of kōura collected, mean CPUE (number of kōura per whakaweku or habitat structure), mean BPUE (biomass of kōura per whakaweku or habitat structure), of kōura collected from tau kōura comprised of 10 whakaweku set at Hiruharama, Motuoapa and nine whakaweku set at Waihi, Lake Taupō, December 2014 and Waihi 2015.

Site	Date	Number of kōura (n)	Mean CPUE (SD)	Mean BPUE (SD)
Hiruharama	17 Dec 2015	234	23.4 (13.5)	622.4 (374.6)
Motuoapa	17 Dec 2015	391	39.1 (16.5)	663.6 (323.7)
Waihi	17 Dec 2015	258	28.7 (18.5)	386.7 (291.8)
Waihi	24 Feb 2015	169	18.8 (15.5)	89.9 (137.7)

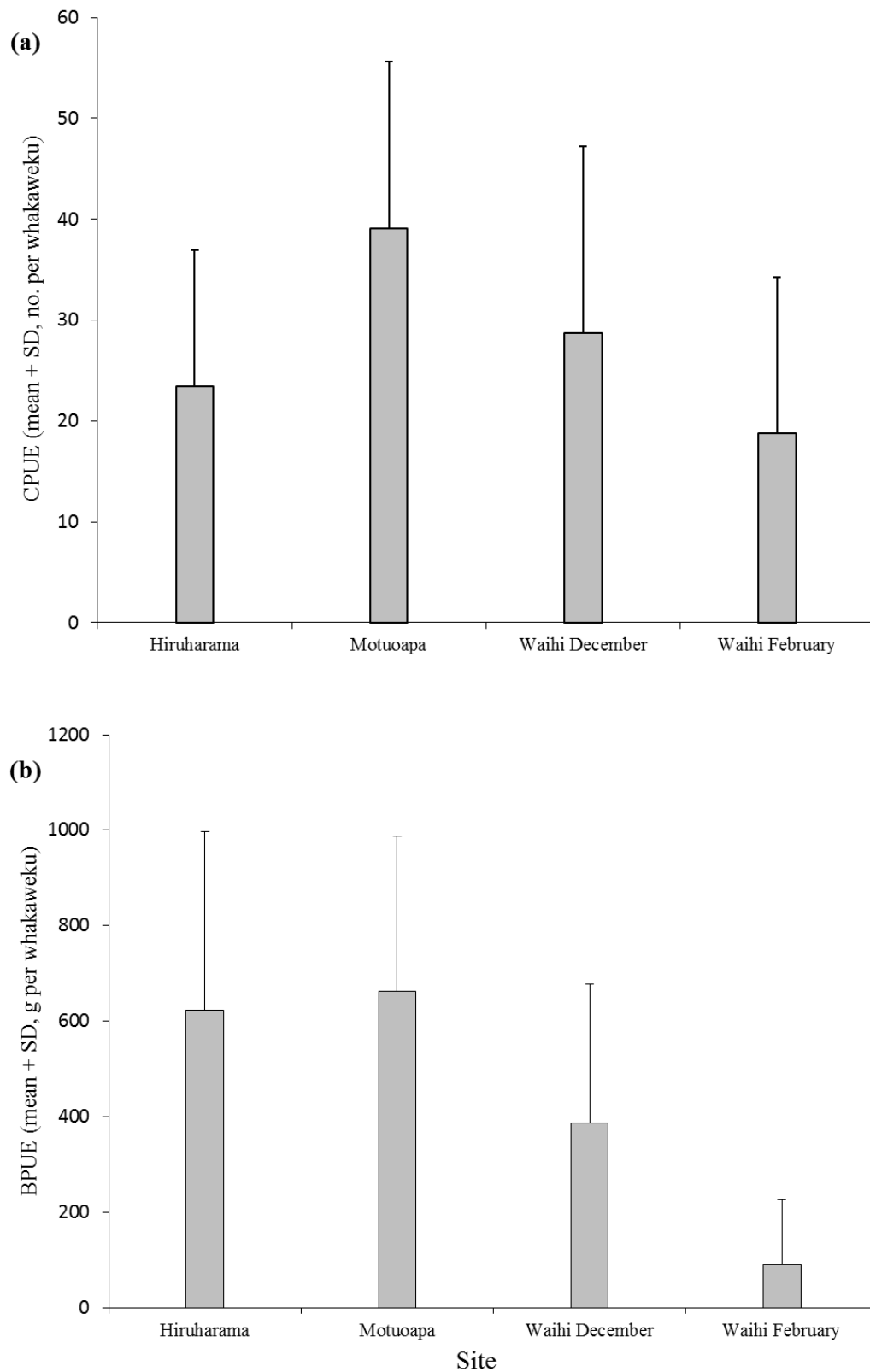


Figure 6 (a) Mean catch per unit effort (CPUE + SD) and (b) mean biomass per unit effort (BPUE + SD) of kōura captured at Hiruharama, Motuoapa and Waihi, Lake Taupō. Tau kōura were set on 22 October 2014 and retrieved from Hiruharama and Motuoapa on 17 December 2014 and from Waihi on 17 December 2014 and 24 February 2015.

4.3 Kōura size

Subsamples of kōura ($n = 476$) were collected and analysed from tau kōura set at Hiruharama ($n = 191$), Motuoapa ($n = 199$) and Waihi ($n = 146$) (Table 3). The highest mean OCL of 30.3 mm recorded at Hiruharama (Table 3). Kōura were smaller at Motuoapa and Waihi with mean OCLs of 25.6 mm and 25 mm, respectively (Table 3; Fig. 7). Kōura OCL ranged from 10 mm to 46 mm at Motuoapa, 12 mm to 46 mm at Hiruharama and 10 mm to 45 mm at Waihi (Table 3). Age of kōura cohorts could not be estimated from length-frequency distributions because shallow habitats, where the majority of smaller sized individuals were found, were under sampled. However, a first-year age class of 10 to 18 mm OCL was identified from the length-frequency graphs for Motuoapa and Hiruharama kōura (Fig. 7).

Table 3 Sampling site, date, number of kōura analysed, percentage female, percentage breeding females, percentage soft shelled, kōura collected from Hiruharama, Motuoapa and Waihi bays, Lake Taupō 17 December 2014 and Waihi Bay 24 February 2015.

Site	Date	Kōura analysed (n)	Mean OCL ($\pm$ SD mm)	OCL range (mm)	Female %	Breeding females %	Soft shells %
Hiruharama	Dec 2014	131	30.3 (8.6)	12 - 46	45.0	11.6	10.7
Motuoapa	Dec 2014	199	25.6 (9.4)	10 - 46	50.3	30.5	5.0
Waihi	Dec 2014	146	25.0 (7.2)	10 - 42	48.3	14.9	2.7
Waihi	Feb 2015	115	23.3 (7.6)	10 - 45	55.4	0	3.5

4.4 Female to male ratio

The overall percentage of females to males was about 1:1, ranging from 45% at Hiruharama to 50.3% at Motuoapa (Table 3). Female kōura were more numerous than males at Waihi in February where they comprised 55.4% of subsamples analysed compared to 48.3% in December (Table 3).

4.5 Soft shelled kōura

Kōura with soft shells were present on all sampling occasions with the highest percentage recorded at Hiruharama (10.7%) in December 2014 (Table 3). The percentage of soft shelled kōura in samples was lower at both Motuoapa (5.0%) and Waihi (2.7%).

4.6 Breeding female kōura

Females with eggs or young were present at all three sites in December, with the highest percentage of breeding size female kōura (defined as >21 mm OCL) recorded at Motuoapa (30.5%) followed by Waihi (14.9%) and Hiruharama (11.6%) (Table 3). No egg-bearing females were captured at Waihi in February (Table 3). Most kōura breed in autumn (late April to early May) and carry eggs over winter.

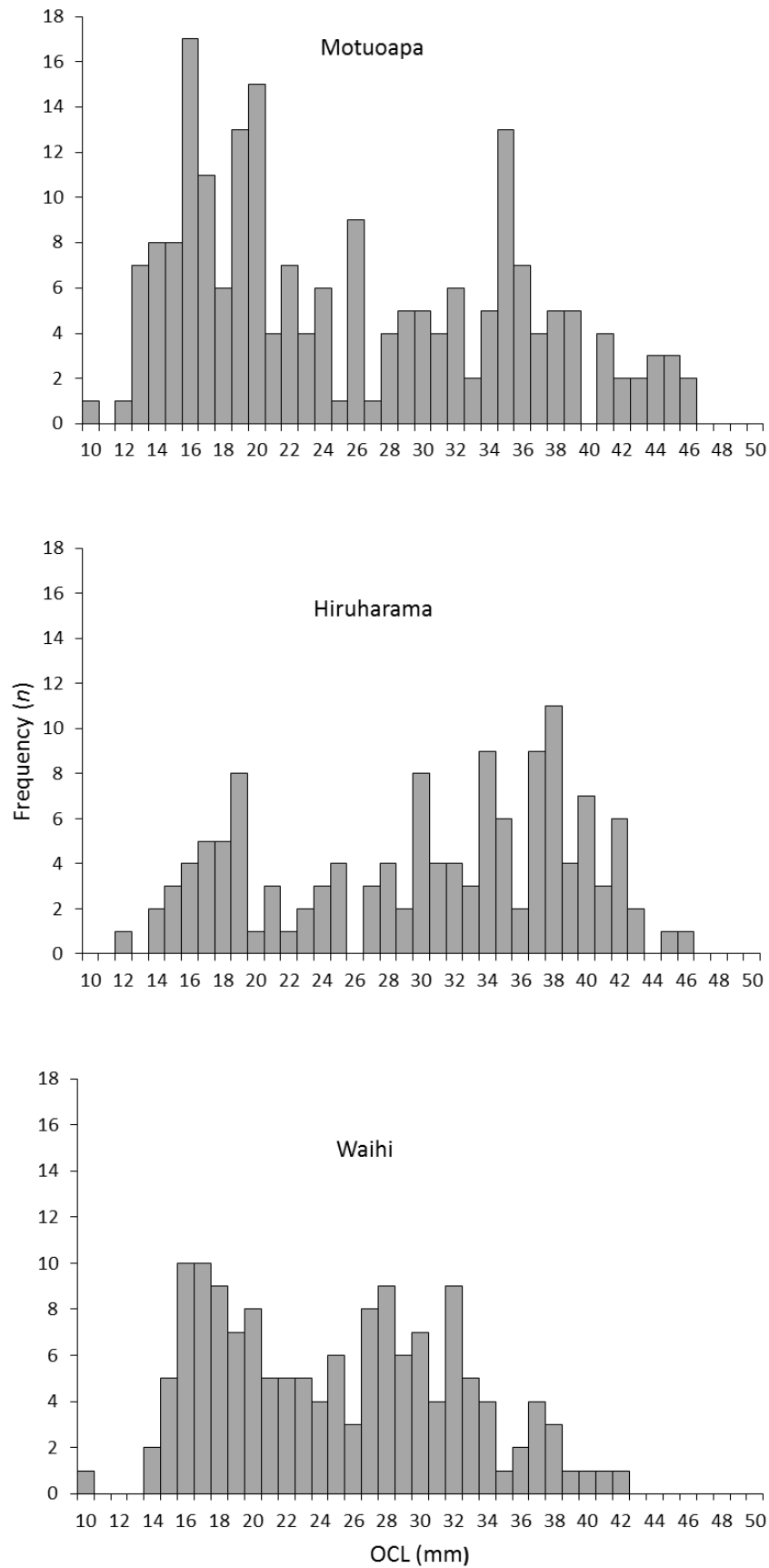


Figure 7 Length (OCL) frequency graph for subsamples of kōura captured from Motuoapa ($n = 200$), Hiruharama ($n = 131$) and Waihi ($n = 146$) Lake Taupō on 17 December 2014.

4.7 Comparison with other lake kōura studies

Kōura abundance, biomass and size estimates from this study were compared with those obtained from the Pukawa and Hiruharama¹ bays in Lake Taupō in 2008 (Kusabs 2008) and from three deep lakes in the Te Arawa lakes district, lakes Rotoiti, Rotomā and Tarawera, in November 2009 (Kusabs, *et al.* 2015b).

CPUE and BPUE of Pukawa and Tarawera kōura were significantly less ($p < 0.05$) than those recorded at Hiruharama (2008 and 2014 surveys), Motuoapa and Waihi and in lakes Rotoiti and Rotomā (Table 4, Fig. 8). Furthermore, Pukawa kōura were significantly smaller ($p < 0.05$) than those at Hiruharama, Motuoapa and Waihi and in lakes Rotoiti and Rotomā. The low abundance of kōura at Pukawa and Tarawera is most probably due to the fine, muddy lake bed substrates present at these sites (Kusabs 2008; Kusabs, *et al.* 2015b) as kōura generally prefer coarse rather than fine sediment (Kusabs, *et al.* 2015b). Given the strong influence of fine sediments on kōura numbers Pukawa and Tarawera were excluded from the inter-lake comparisons. The subsequent analyses showed no significant differences in kōura CPUE ($p > 0.1$) or BPUE ($p > 0.1$) between lakes Rotoiti, Rotomā and Taupō (Pukawa excluded) (Table 4, Fig. 8). Taupō kōura were significantly smaller ($p < 0.05$) than those in Rotomā but not Rotoiti (Table 4).

Table 4 Sampling site, date, number of kōura analysed, mean CPUE (number of kōura per whakaweku or habitat structure), mean BPUE (biomass of kōura per whakaweku), mean size (OCI; mm) of kōura collected from Hiruharama, Pūkawa, Motuoapa and Waihi bays, Lake Taupō 2005 to 2008. Hiruharama and Pūkawa Bays data from Kusabs (2008). Data for Rotoiti, Rotomā and Tarawera from Kusabs, *et al.* (2015a).

Site	Date	Mean CPUE (n ± SD)	Mean BPUE (g ± SD)	Mean OCI (mm ± SD)
Hiruharama	Nov 2008	35.0 (3.5)	856.5 (313.1)	30.1 (8.4)
Hiruharama	Dec 2015	23.4 (13.5)	622.0 (374.6)	30.3 (8.6)
Pūkawa	Nov 2008	6.9 (4.6)	69.0 (64.7)	21.5 (6.1)
Motuoapa	Dec 2015	39.1 (16.5)	664.0 (323.7)	25.6 (9.4)
Waihi	Dec 2015	28.7 (18.5)	386.7 (137.7)	25.0 (7.2)
Rotoiti	Nov 2009	23.2 (15.4)	443.6 (61.8)	28.7 (5.8)
Rotomā	Nov 2009	26.2 (18.7)	742.4 (136.1)	31.3 (7.7)
Tarawera	Nov 2009	4.2 (4.1)	69.8 (16.3)	25.0 (8.5)

¹ Tau kōura set at two different locations - refer Fig.1.

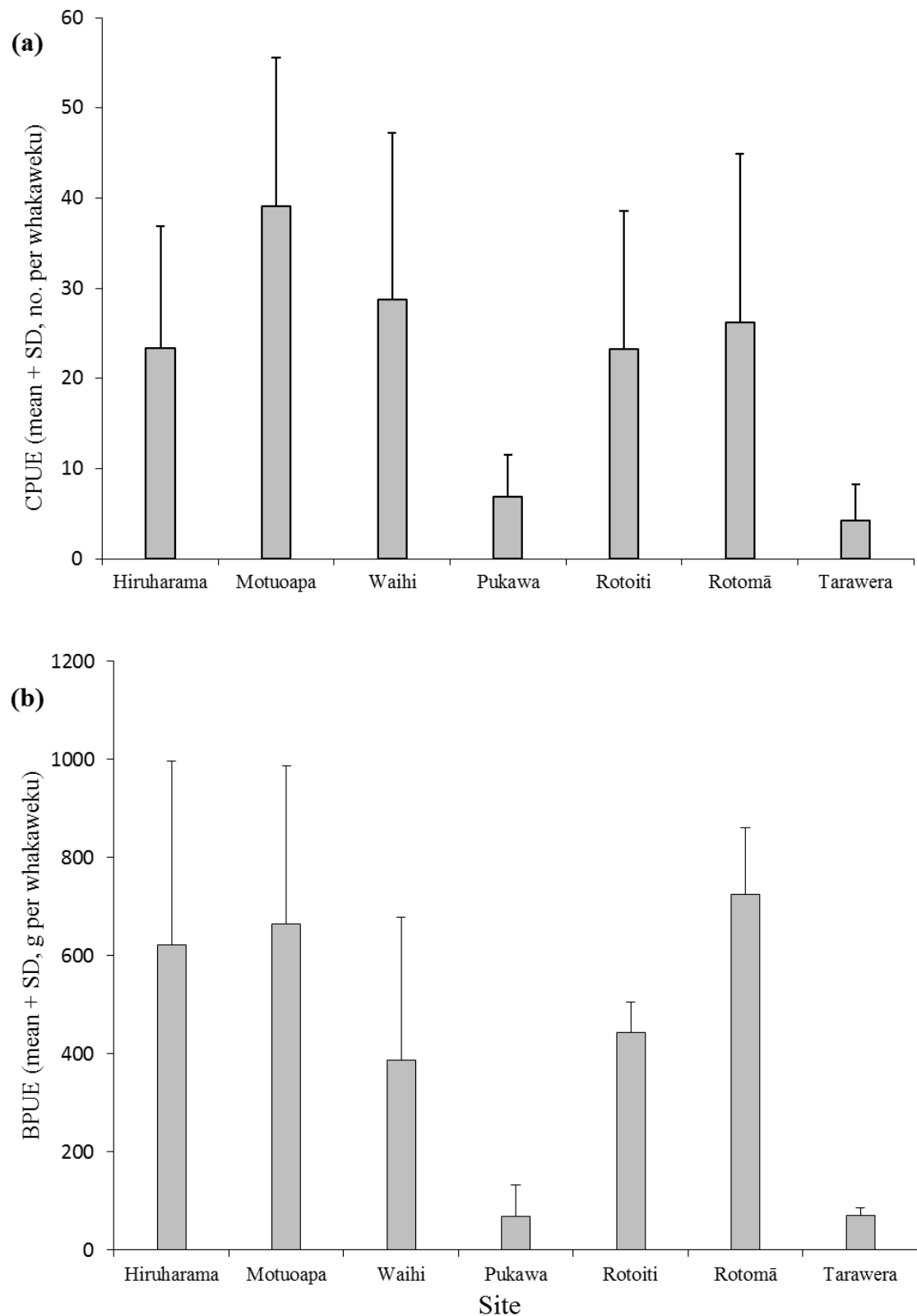


Figure 8 (a) Mean catch per unit effort (CPUE + SD) and (b) mean biomass per unit effort (BPUE + SD) of kōura captured at Hiruharama, Motuoapa, Waihi and Pukawa, Lake Taupō and lakes Rotoiti, Rotomā and Tarawera. Hiruharama, Motuoapa and Waihi December 2014 data from this study; Pukawa data, November 2008, from (Kusabs 2008) and lakes Rotoiti, Rotomā and Tarawera data for November 2009 from Kusabs, *et al.* (2015b).

4.8 Catfish

Catfish were abundant in Lake Taupō from 1997 – 2015 the mean CPUE was 34 catfish per net² and has varied from 8.1 catfish per net (2004/05) to 87 catfish per net (1999/2000) (Fig. 9, Appendix 1). Male catfish appear to be larger than females with a mean FL³ of 232 mm and a mean weight⁴ of 191 g compared to a mean FL of 217 mm and a mean weight of 161 g (Fig. 10, Appendix 2). From a precursory inspection of the data the catfish population appears to have declined since 2000 and is now composed of fewer, but larger individuals (Figs. 9 & 10). A detailed analysis of the original data set is required to confirm whether this trend is real or perceived.

Dietary analysis of catfish collected from Waihi and Motuoapa during the 2014/15 sampling season (November 2014, to February 2015) showed that catfish consume mainly vegetation and snails. Kōura comprised a small component of the diet, occurring in the stomach contents of only two catfish from a total sample of 283 fish (Department of Conservation records). This most likely reflects the type of habitat sampled (i.e. shallow, sandy and weedy) as it is probable that catfish still consume large amounts of kōura in rocky habitats as reported by Barnes (1996).

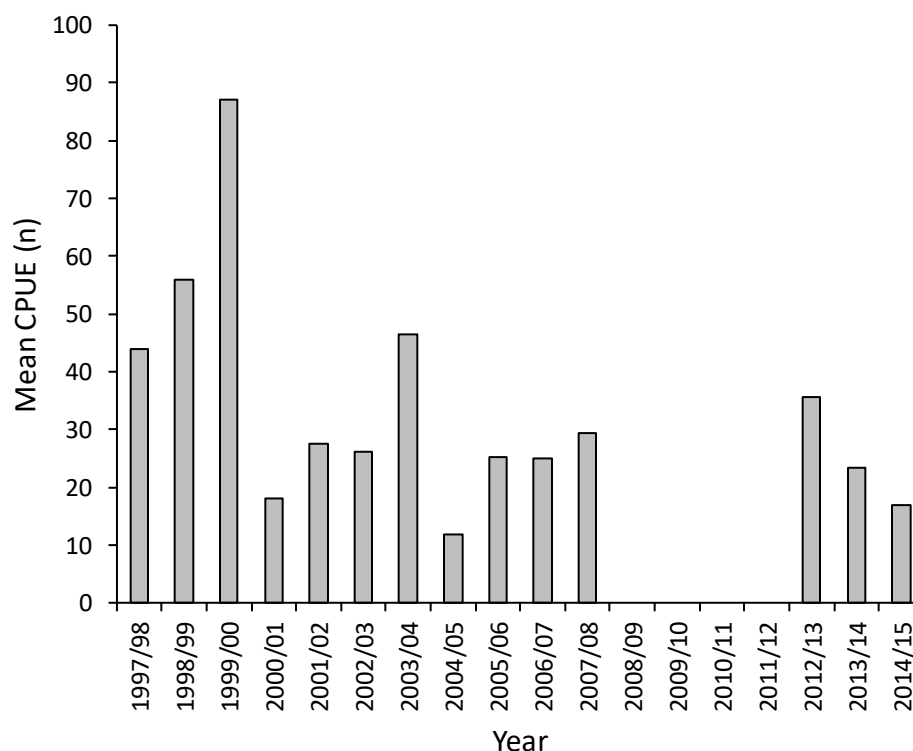


Figure 9. Mean catch per unit effort (CPUE) of catfish captured in Lake Taupō, 1997 to 2013. Note: no data available for 2008-2010. Data courtesy of Department of Conservation.

² Mean of means for the period 1997 to 2015.

³ Mean of mean FL's, 1997-2014

⁴ Mean of mean weights 1996-2014

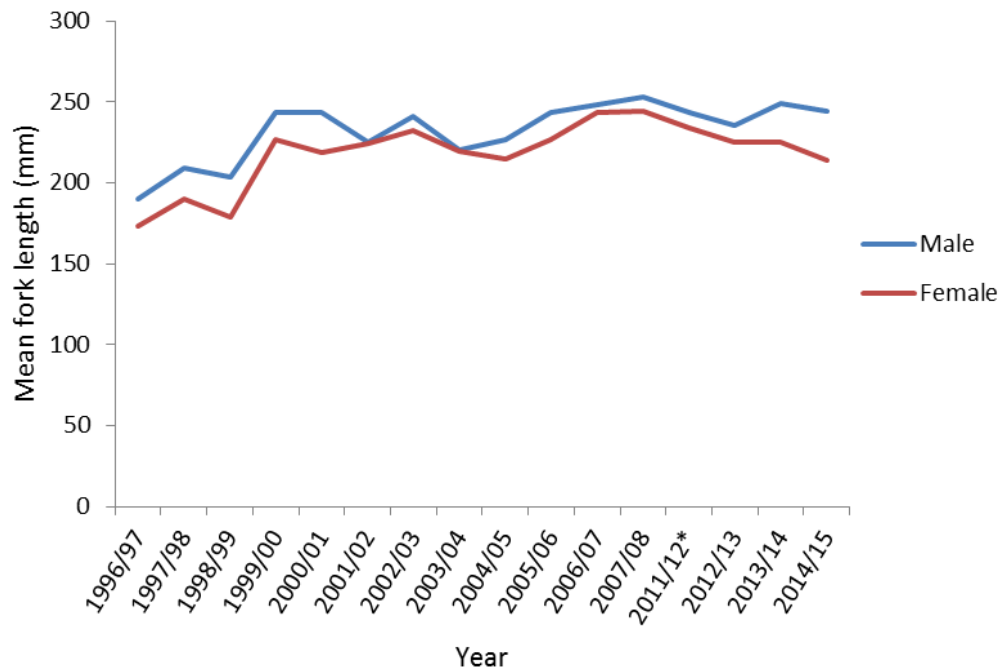


Figure 10 Mean fork length (mm) of male and female catfish in Lake Taupō, 1996 to 2014. Note: no data available for 2008-2010. Data courtesy of Department of Conservation.

5 DISCUSSION

5.1 Kōura abundance in Lake Taupō and the importance of lakebed substrates

High numbers of harvestable sized kōura (>28 mm OCL) were found at the three study sites, Motuoapa, Hiruharama and Waihi, in Lake Taupō. Kōura abundance, biomass and size of Taupō kōura were similar to those recorded for kōura populations in lakes Rotoiti and Rotomā where catfish were not present.

Kōura populations were significantly more abundant and had a higher mean size at Motuoapa, Hiruharama and Waihi than at Pukawa or at Lake Tarawera. This is most likely due to the muddy lake bed substrates present at these locations and suggests that sediment particle size is the strongest driver of kōura populations in Lake Taupō. This is consistent with studies in the Te Arawa lakes and overseas which have generally identified substrate as the most important variable relating to crayfish abundance (Capelli and Magnuson 1983; Kirjavainen and Westman 1999; Hein, *et al.* 2006) irrespective of lake trophic status or predatory fish density (Nyström, *et al.* 2006; Kusabs, *et al.* 2015b).

Cobble substrates are particularly important to lake crayfish as they provide shelter from predators (Stein 1977) and also prevent cannibalism of egg-bearing females and of individuals undergoing moult (Stein 1977; Capelli and Magnuson 1983; Kershner and Lodge 1995; Nyström, *et al.* 2006). Because of their small size and softer exoskeleton, juvenile crayfish is the life stage most vulnerable to predatory

fish, but as crayfish grow they are less likely to be consumed by these gape size-limited predators (Stein and Magnuson 1976; Didonato and Lodge 1993). Even in the absence of benthic predatory fish kōura generally prefer coarse rather than fine sediment. Devcich (1979) reported very low numbers of kōura ($<0.01 \text{ m}^{-2}$) on soft muddy sediments deeper than 23 m in Lake Rotoiti, North Island. Similarly, in the Waikato River hydro lakes, kōura were commonly associated with well-consolidated substrata, such as, rock crevices, boulders and submerged trees, but were rarely found on muddy sediments (Coffey 1995).

5.2 The effects of catfish on kōura populations in Lake Taupō

Studies of catfish diet in Lake Taupō show that they regularly consume kōura (Barnes and Hicks, 2003; Department of Conservation data), however the effect of catfish on overall kōura abundance in Lake Taupō may be limited because there is little direct overlap in the habitat distributions of the two species. In Lake Taupō, catfish are concentrated in the littoral zone and macrophyte beds whereas kōura are most abundant at rocky sites and at depths outside of the macrophyte beds.

Dedual (2002) used ultrasonic telemetry to investigate the vertical distribution and movement of catfish in Motuoapa Bay reported that catfish generally swim within the top 10 m of water and occasionally in water as deep as 17 m. Catfish were found to favour the presence of rooted plants, therefore they were mainly concentrated in the littoral zone where weed beds were present (Dedual 2002). In our study, tau kōura were set at depths ranging from 9.5 m to 33 m outside the depths ($<10 \text{ m}$) commonly inhabited by catfish. Therefore, the shallow habitats where catfish and kōura interactions are likely to be greatest were under sampled. However, this did not affect our inter-lake comparisons as tau kōura in lakes Rotomā, Rotoiti and Tarawera were also deployed outside of the depth range of the tall growing aquatic macrophyte beds (Fig. 1).

The interaction between kōura and macrophyte beds in New Zealand lakes is poorly understood however invasive macrophytes are known to establish dense stands that directly interfere with movement of adult crayfish and cover shallow areas of lakes, thereby, excluding crayfish (Hessen, *et al.* 2004). Lake Steinsfjorden (southeast Norway) is the only well studied case of an exotic macrophyte invasion on a native crayfish population, *Astacus astacus* (Skurdal and Qvenild 1989; Skurdal, *et al.* 2002; Hessen, *et al.* 2004). Canadian pondweed *Elodea canadensis* was for the first time observed in the lake in 1977. Over the following years, the plant established dense cover over large parts of the shallow areas, excluding the crayfish from these areas and causing a sudden drop in population size (i.e., from 194,000 to 92,000 harvestable sized crayfish) (Hessen, *et al.* 2004). It is probable that exotic macrophytes, in conjunction with catfish predation, have similarly excluded kōura from shallow

habitats in Lake Taupō. Further research into the abundance and distribution of kōura in the littoral zone and macrophyte beds in Lake Taupō is required to verify this hypothesis.

Barnes (1996) found catfish to be more abundant in the southern bays, including Waihi Bay and Motuoapa Headland, than in the exposed eastern shores and northern harbours in Lake Taupō. Therefore, it would be expected that the effect of catfish on kōura populations would be most evident in the southern bays. However, our results show that kōura are locally abundant at Motuoapa and Waihi (at depths from 17 m to 33 m) and that kōura abundance and biomass at these sites were not significantly different from Hiruharama at the northern end of the lake. Moreover, Department of Conservation monitoring data suggest that the Lake Taupō catfish population has stabilised since their illegal introduction in the 1980's and is now composed of fewer but larger fish. Barnes and Hicks (2003) suggested that the limited areas where macrophytes are able to establish within Lake Taupō may be an important factor controlling the reproductive potential of catfish, preventing the population from reaching high levels throughout the lake.

5.3 Interference with tau kōura

The tau kōura is an effective method of monitoring kōura however interference and theft of tau kōura lines is a major problem in Lake Taupō. In this study, it was intended to carry out spring and summer surveys and following the completion of these surveys that the tau kōura would be left in place for future monitoring purposes and for use by local hapu. Unfortunately, the tau kōura at Hiruharama was removed completely while at Motuoapa the tau kōura was cut in half. This has occurred previously at Hiruharama in 2008 and also in lakes Rotomā and Tarawera in the Te Arawa lakes district. Interference is more common in oligotrophic lakes as tau kōura lines are more easily detected in the clear waters. Tau kōura can 'snag' angling tackle and this can result in anglers lifting the tau kōura to retrieve their lures. If tau kōura surveys are to be carried out in Lake Taupō in the future then more effort will be needed to protect them from interference. Potential measures may include; locating tau kōura more cryptically in low-use sites, closing off areas of the lake where tau kōura are sited and increasing public awareness.

6 CONCLUSIONS

Kōura remain locally abundant in Lake Taupō despite the presence of catfish, with high numbers of harvestable sized kōura present at the three study sites, Motuoapa, Hiruharama and Waihi. Kōura abundance and biomass estimates in Lake Taupō were similar to those recorded at lakes Rotoiti and Rotomā where catfish are absent.

It is possible that catfish have had a limited effect on overall kōura abundance in Lake Taupō because there is minimal habitat overlap with catfish concentrated in the littoral zone and macrophyte beds (<10 m) and kōura most abundant at rocky sites and at depths outside of the macrophytes (>10 m). Exotic macrophytes generally provide poor habitat for kōura and it is likely that dense macrophyte beds, in conjunction with catfish predation, have excluded kōura from shallow areas in Lake Taupō. However, our study suggests that catfish may have had a limited impact on kōura populations at depths outside of the macrophytes.

The results of this study are consistent with studies in the Te Arawa lakes and overseas which have generally identified lakebed substrate as the most important variable relating to crayfish abundance irrespective of lake trophic status or predatory fish density (Nyström, *et al.* 2006; Kusabs, *et al.* 2015b). Cobble substrates in Lake Taupō may be particularly important as they provide shelter for juvenile kōura, the most vulnerable life stage, from predators (i.e. catfish and trout).

The tau kōura is an effective method of monitoring kōura in Lake Taupō, however, vandalism and theft is a major issue. If tau kōura are to be deployed in Lake Taupō in the future then measures will have to be implemented to protect them from human interference.

7 REFERENCES

- Barnes, G.E. 1996. The biology and general ecology of the brown bullhead catfish (*Ameiurus nebulosus*) in Lake Taupō. Unpublished MSc thesis, University of Waikato, Hamilton, New Zealand.
- Barnes, G.E. and Hicks, B.J. 2003. Brown bullhead catfish (*Ameiurus nebulosus*) in Lake Taupō. Managing invasive freshwater fish in New Zealand. Proceedings of a workshop hosted by Department of Conservation. Hamilton, New Zealand. pp.27-35.
- Capelli, G.M. and Magnuson, J.J. 1983. Morphoedaphic and biogeographic analysis of crayfish distribution in Northern Wisconsin. *Journal of Crustacean Biology* **3**, 548-564.
- Coffey, B.T. 1995. Lake Ohakuri- Middle Waikato River - a contribution to a biological resource inventory for resident, submerged, macro-organisms. Report prepared for the Waikato Regional Council. Brian T. Coffey and Associates Limited, Hamilton, New Zealand.
- Dedual, M. 2002. Vertical distribution and movements of brown bullhead (*Ameiurus nebulosus* Lesueur 1819) in Motuoapa Bay, southern Lake Taupō, New Zealand. *Hydrobiologia* **483**, 129-135.
- Devcich, A.A. 1979. An ecological study of *Paranephrops planifrons* (White) (Decapoda: Parastacidae) in Lake Rotoiti, North Island, New Zealand. PhD, University of Waikato, Hamilton, New Zealand.
- Didonato, G.T. and Lodge, D.M. 1993. Species replacements among Orconectes crayfishes in Wisconsin lakes - the role of predation by fish. *Canadian Journal of Fisheries and Aquatic Sciences* **50**, 1484-1488.
- Fletcher, J.H. 1919. The edible fish of Taupo-nui-a-Tia. *Transactions of the New Zealand Institute* **51**, 259-264.

- Hein, C.L., Roth, B.M., Ives, A.R. and Zanden, M.J.V. 2006. Fish predation and trapping for rusty crayfish (*Orconectes rusticus*) control: a whole-lake experiment. *Canadian Journal of Fisheries and Aquatic Sciences* **63**, 383-393.
- Hessen, D., Skurdal, J. and Braathen, J. 2004. Plant exclusion of a herbivore; crayfish population decline caused by an invading waterweed. *Biological Invasions* **6**, 133-140.
- Jolly, V.H. 1968. The comparative ecology of some New Zealand lakes 1. physical and chemical. *New Zealand Journal of Marine and Freshwater Research* **2** (2), 214-259.
- Kershner, M.W. and Lodge, D.M. 1995. Effects of littoral habitat and fish predation on the distribution of an exotic crayfish, *Orconectes rusticus*. *Journal of the North American Benthological Society* **14**, 414-422.
- Kirjavainen, J. and Westman, K. 1999. Natural history and development of the introduced signal crayfish, *Pacifastacus leniusculus*, in a small, isolated Finnish lake, from 1968 to 1993. *Aquatic Living Resources* **12**, 387-401.
- Kusabs, I. and Butterworth, J. 2013. Kōura abundance and distribution in Lake Rotoehu. Report prepared for Bay of Plenty Regional Council. Ian Kusabs & Associates Ltd, Rotorua, New Zealand.
- Kusabs, I., Emery, W. and Butterworth, J. 2014. Ohau Channel Diversion Wall - An assessment of the kōura and kākahi populations in the Okere Arm and Lake Rotoiti. Report prepared for Bay of Plenty Regional Council. Ian Kusabs & Associates Ltd, Rotorua, New Zealand.
- Kusabs, I., Emery, W., Parkyn, S., Hickey, C. and Quinn, J. 2005. Traditional method a winner for monitoring lake kōura populations. Available from http://www.niwa.co.nz/our-science/te-kuwaha/research-projects/all/monitoring-koura/monitoring_koura Accessed [July 2014].
- Kusabs, I. and Taiaroa, R. 2015. Kōura and catfish populations in Lake Taupō - literature review. Report prepared for Ngāti Tūwharetoa Māori Trust Board. Ian Kusabs & Associates Ltd, Rotorua, New Zealand.
- Kusabs, I.A. 2008. A preliminary assessment of Kōura populations in Lake Taupō. Report prepared for Tūwharetoa Asset Management Unit. Ian Kusabs and Associates Ltd, Rotorua, New Zealand. 11pp.
- Kusabs, I.A., Hicks, B.J., Quinn, J.M. and Hamilton, D.P. 2015a. Sustainable management of freshwater crayfish (kōura, *Paranephrops planifrons*) in Te Arawa (Rotorua) lakes, North Island, New Zealand. *Fisheries Research* **168**, 35-46.
- Kusabs, I.A. and Quinn, J.M. 2009. Use of a traditional Māori harvesting method, the tau kōura, for monitoring kōura (freshwater crayfish, *Paranephrops planifrons*) in Lake Rotoiti, North Island, New Zealand. *New Zealand Journal of Marine and Freshwater Research* **43**, 713-722.
- Kusabs, I.A., Quinn, J.M. and Hamilton, D.P. 2015b. Effects of benthic substrate, nutrient enrichment and predatory fish on freshwater crayfish (kōura, *Paranephrops planifrons*) population characteristics in seven Te Arawa (Rotorua) lakes, North Island, New Zealand. *Marine and Freshwater Research* -.
- McDowall, R.M. 1990. *New Zealand freshwater fishes - a natural history and guide*. Heinemann Reed Publishing (NZ) Ltd, Auckland.
- Nyström, P., Stenroth, P., Holmqvist, N., Berglund, O., Larsson, P. and Granéli, W. 2006. Crayfish in lakes and streams: individual and population responses to predation, productivity and substratum availability. *Freshwater Biology* **51**, 2096-2113.
- Parkyn, S., Kusabs, I., De Winton, M. and Taumoepeau, A. 2006. Kōura aquaculture in Lake Taupo: use of harvestable habitat refuges. NIWA Client Report: HAM2006-170. National Institute of Water and Atmospheric Research Ltd, Hamilton, New Zealand.
- Skurdal, J., Garnas, E. and Taugbol, T. 2002. Management strategies, yield and population development of the noble crayfish *Astacus astacus* in Lake Steinsfjorden. *Bulletin Francais De La Peche Et De La Pisciculture* 845-860.

- Skurdal, K.J. and Qvenild, T. 1989. *Astacus astacus* in Lake Steinsfjorden after Canadian waterweed *Elodea canadensis* invasion. *Freshwater Crayfish* **7**, 47-52.
- Stein, R.A. 1977. Selective predation, optimal foraging, and predator-prey interactions between fish and crayfish. *Ecology* **58**, 1237-1253.
- Stein, R.A. and Magnuson, J.J. 1976. Behavioral response of crayfish to a fish predator. *Ecology* **57**, 751-761.
- Vincent, W.F. and Forsyth, D.J. 1983. Plankton of the open water. *In* Lake Taupo: Ecology of a New Zealand lake. Department of Science and Research Information Series 158. *Edited by* D. J. H.-W. Forsyth, C. Department of Science and Research, Wellington, New Zealand.
- Viner, A.B. and White, E. 1987. Phytoplankton growth. *In* Inland waters of New Zealand. Department of Science and Research Bulletin 241. *Edited by* A. B. Viner. Department of Science and Research, Wellington, New Zealand. pp.191-223.

8 Appendices

Appendix 1 Mean CPUE of catfish in Waihi, Motuoapa and Whakaipo Bays, Lake Taupō, 1997 to 2014. Number of fyke nets used ranged between 2 and 6 per site. Data courtesy of Department of Conservation.

Year	Months sampled	Mean CPUE (catfish per net)
1997/98	Nov, Dec, Jan, Feb, Mar	44.0
1998/99	Dec, Jan, Feb, Mar	56.0
1999/00	Nov, Dec, Jan, Feb, Mar	87.0
2000/01	Nov, Jan, Mar	18.1
2001/02	Nov, Jan, Mar	27.6
2002/03	Nov, Jan, Mar	26.2
2003/04	Nov, Jan, Mar	46.5
2004/05	Nov, Jan, Mar	11.8
2005/06	Nov, Dec, Jan, Feb, Mar	25.3
2006/07	Nov, Dec, Jan, Feb, Mar	25.1
2007/08	Nov, Dec, Jan, Feb, Mar	29.3
2008/09	-	-
2009/10	-	-
2010/11	-	-
2011/12	-	-
2012/13	Nov, Dec, Jan, Feb, Mar	35.6
2013/14	Nov, Dec, Jan, Feb, Mar	23.3
2014/15	Nov, Dec, Jan, Feb, Mar	17.0

Appendix 2 Mean fork length, weight of catfish in Waihi, Motuoapa and Whakaipo Bays, Lake Taupō, 1996 to 2014. Number of fyke nets used ranged between 2 and 6 per site. Data courtesy of Department of Conservation. * netting carried out in Waihi Bay only.

Year	<u>Male</u>		<u>Female</u>		<u>Immature</u>	
	Mean length (mm)	Mean weight (g)	Mean length (mm)	Mean weight (g)	Mean length (mm)	Mean weight (g)
1996/97	189.8	113.6	173.6	84.7	119.4	22.7
1997/98	209.2	149.7	189.7	117.1	130.9	31.1
1998/99	203.4	142.5	178.6	91.6	133.3	38.9
1999/00	243.5	218.4	226.6	169.5	135.6	44
2000/01	243.5	230	218.9	180.5	130.8	46.6
2001/02	225.4	159.4	224.1	154.7	126.3	24
2002/03	241.1	200.7	232	189.3	125.6	23.6
2003/04	220.6	159.4	219.5	161.7	120	20.3
2004/05	226.7	198	214.8	172.6	142.5	41.4
2005/06	243.3	217.8	226.9	182.7	125	28.1
2006/07	248.2	216.7	243.6	217.5	250	185
2007/08	252.9	222.7	243.9	204.9	159	66.6
2008/09	-	-	-	-	-	-
2009/10	-	-	-	-	-	-
2010/11	-	-	-	-	-	-
2011/12*	243.1	195.2	234.1	177.7	-	-
2012/13	235.8	192.5	225.0	170.6	117.8	22.5
2013/14	249.0	223.8	225.0	171.8	131.2	41.2
2014/15	244.2	221.5	214.1	144.9	113.5	20.5